R documentation

of 'NEWS.Rd'

October 11, 2026

NEWS *News for Package* **DirichletReg**

Changes in Version 0.8-0

- Bumped up the required R-version to 4.1.0.
- Fixed `print.DirichletRegModel` which erroneously issued an error (convergence) when **maxLik**'s `maxNR` returned convergence-code 8 (see `help("maxNR", package = "maxLik")`; thanks to Nick Thieme). Instead of stopping with an error, a warning is now issued with more detailed diagnostic information.
- `DirichReg()` was completely rewritten and now handles formula input much more robustly and predictably. Aside from some hopefully helpful new messages, there should not be any user-visible changes.
- `DR_data()` was rewritten to address some issues in the original implementation. `trafo` and `norm_tol` must now be $\in (0, .1]$. `norm_tol` must be numeric but `trafo` can also be `TRUE` to force transformation even if it is not necessary (not `FALSE` because this would lead to errors when estimating the models). If the data are supplied as a single column or vector, the function tries to guess and fix the new column names if `fix_names = TRUE`. NB: The attribute `attr(*, "Y.original")` now stores the original data `as.matrix`-transformed.
- `DirichReg()` now evaluates the `formula` object in the parent frame to ensure that character-to-formula and similar applications can be processed correctly (see https://github.com/maiermarco/DirichletReg/issues/4).
- Added a `coef` method for objects of class `summary_DirichletRegModel` generated by the `summary` method (see https://github.com/maiermarco/DirichletReg/issues/5).
- C-Routines are now symbols prefixed with `C_` (e.g., `C_wght_LL_grad_alternative`).
- `.onUnload` now also unloads the DLL.
- Using **revdepcheck** to see if changes break other packages on CRAN or BioConductor.
- Changed call to `structure()` from the deprecated special name `.Dim` to `dim`.
- Fixed a bug in `DirichReg` that prevented users from supplying starting values via `control = list(sv = c(...))`. An example of how to use it has been added to `?DirichReg`.
- Rewrote the `update.DirichletRegModel` method to address issues with scoping.
- Verbosity now defaults to `getOptions("verbose")` if not specified otherwise.

- General cleanup of the package (old CI-integrations, citations, ...). Unused old unexported functions are now deprecated with a warning and will be removed in subsequent versions.
- Revision of the documentation.
- Better code documentation.
- Adapted the tests for the third edition of **testthat**.
- Many small improvements suggested by **lintr**.
- Added `[.DirichletRegData` and `[<-.DirichletRegData` methods.
- Added new return codes from **maxLik**.

Changes in Version 0.7-2 (2025–05–31)

- Implemented CRAN-Notes.

Changes in Version 0.7-1 (2021–04–29)

- `summary.DirichletRegModel` now returns a `summary_DirichletRegModel`-class object without printing it (see https://github.com/maiermarco/DirichletReg/issues/3).
- Moved **rmarkdown** to Suggests to comply with new version of **knitr** (see https://github.com/yihui/knitr/issues/1864).

Changes in Version 0.7-0 (2020–05–28)

- Fixed: Cutoff in formula-deparsing removed.
- Removed **rgl** dependency (package will be loaded if necessary).
- Moved to GitHub.

Changes in Version 0.6-3 (2015–11–12)

- Instead of producing an error, density functions now return `NaN` with a warning if any element in `alpha` is <= 0.

Changes in Version 0.6-2 (2015–02–09)

- Fixed an error in the ternary plot (bottom axis tick labels were printed in reversed order; pointed out by Emilio A. Laca).
- Imports functions from default packages, as required by the new CRAN-check.
- Changed the `Description` field to eliminate the note "Malformed Description field: should contain one or more complete sentences." when checking the package.

Changes in Version 0.6-1 (2014–12–12)

- Re-derived gradients/Hessians, optimized C-code, and tweaked computation of starting values which leads to considerably better performance.
- Fixed some bugs in `drop1`.
- Expanded **testthat** checks.

Changes in Version 0.6-0

- All likelihood and gradient functions are now written in C (all called via `.Call()` now instead of `.C()`) which leads to a considerable gain in speed.
- A `drop1` method for Dirichlet regression models was added. As it is still experimental and will probably change, use it with care. Options such as `scope` will be added in one of the next releases.
- Fixed a bug in the `print` method for confidence intervals and one in `confint.DirichletRegModel()`.
- Expanded the **testthat** check- and test-suite.
- Known issue: If you have collinear (aliased) terms, the estimation will fail. This will be handled automatically in subsequent releases, but for now, please remove the respective terms. If you fit a model and it says something like: `Error in prepareFixed(start = start, activePar = NULL, fixed = fixed): At least one parameter must not be fixed using argument 'fixed'` you most likely have collinear terms or "empty" combinations of interaction terms.

Changes in Version 0.5-2 (2014–09–18)

- Fixed checking functions in `tests/testthat`.

Changes in Version 0.5-1 (2014–09–16)

- Fixed a bug when using the `subset` argument.
- Added tolerance for normalization check to `DR_data`.
- moved `NEWS` to the new fancy `NEWS.Rd` file/format.
- Added the possibility to do quick analyses and transforming data "on the fly", like `DirichReg(DR_data(A) ~ 1)`. However this is only intended for quick checking purposes and may be removed at any time.

Changes in Version 0.5-0

- Transformation in `DR_data` is now not only `TRUE/FALSE`, but, by default, a small numeric value to avoid troubles with floating point numbers close to 0 or 1.
- Time-critical routines were implemented in C (pure R versions are available, see `?ddirichlet`).
- `anova.DirichletRegModel` now invisibly returns results as an object that is printed by a method.
- Optimized estimation routines.
- Fixed a bug in the predict method.
- Started development of a comprehensive test-suite using **testthat**.
- Published a working paper on the package:
  Maier, M. J. (2014). *DirichletReg: Dirichlet Regression for Compositional Data in R.* Research Report Series / Department of Statistics and Mathematics, 125. WU Vienna University of Economics and Business, Vienna. doi:10.57938/ad3142d32fcd4c37aec68e0bd7d077e1
- Added vignette with code to the working paper.
- Added citation info.

Changes in Version 0.4-1

- The `trafo` Argument of `DR_data` has been changed, because it has lead to problems in practical applications when numbers very close to 0 or 1 were present.
- `DR_data` checks for negative values and generates an appropriate error message.
- `DR_data` has been made more robust in the presence of `NA`s.

Changes in Version 0.4-0 (2012–04–02)

- Data structure generated by `DR_data` has changed – the new objects can now be integrated into data frames.
- Formula processing is now handled by the package **Formula**.
- New methods have been implemented, especially for the class `DirichletRegModel`.
- The documentation is now quite complete.
- Some speed improvements could be achieved.
- Lots of minor (invisible) changes.

Changes in Version 0.002

- Added the analytical Gradient and Hessian for both parametrizations.
- Optimization: preliminary results by BFGS that become starting values for Newton-Raphson optimization computing the final results.
- Implemented some residuals
- Updated help entries

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